

Challenge: **Early Detection of Leaks in Drinking Water Pipelines**

(SDG 6, target 6.4)

1 Selection of a Specific SDG Target

Target 6.4 of the Sustainable Development Goals (SDG 6: Clean Water and Sanitation) aims to substantially increase water-use efficiency across all sectors and ensure sustainable freshwater withdrawal and supply to address water scarcity. Within this framework, the early detection of leaks in drinking water distribution networks constitutes a critical problem. Undetected leaks can account for significant losses of distributed water, reaching nearly 40% in some systems. This reduces the effective availability of drinking water, increases operational costs, and intensifies pressure on already limited freshwater resources.

2 Identification of the Computational Bottleneck

Early leak detection in drinking water distribution networks can be performed using pressure sensors that record anomalous variations in the hydraulic network. However, installing sensors at every node is economically infeasible; therefore, the problem consists of identifying the minimum set of sensor locations that maximizes leak diagnosticability by selecting positions with high sensitivity to leak scenarios and low redundancy among them. In other words, the selected locations should exhibit significant pressure changes when leaks occur, while avoiding sensor placements that provide similar information for the same leak scenarios.

Evaluating a sensor configuration requires simulating multiple leak scenarios, since a leak may occur at any point along a pipe and its effect depends on the network topology, hydraulic conditions, and sensor placement. Because the number of possible configurations grows combinatorially with the number of candidate nodes, exhaustive search for the optimal solution becomes computationally expensive for classical methods.

Therefore, the problem is formulated as a combinatorial optimization problem on the network: a binary variable indicates whether a sensor is installed at each candidate node, while the objective function maximizes leak diagnosability while penalizing the total number of sensors. This discrete structure makes the problem mathematically tractable and suitable for exploring quantum optimization approaches.

3 Simulated Test Data

To evaluate the proposed algorithms, a dataset generated through numerical simulations of a drinking water network will be provided. This example is intended to enable an initial assessment of candidate solutions; however, the developed algorithm should be general and applicable to networks with different topologies, sizes, and hydraulic conditions.

The network topology is represented by a graph in which nodes correspond to potential locations for pressure sensors and edges represent the pipelines of the network.

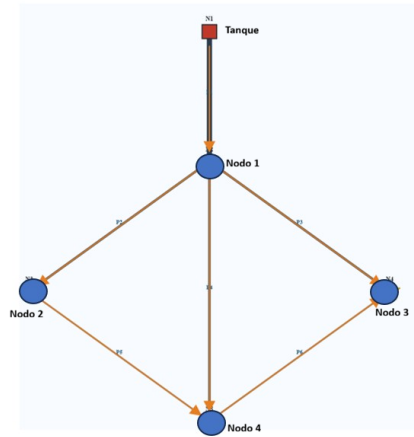


Figure 1: Topology of the simulated potable water network[5].

Based on this topology, the nominal (leak-free) behavior of the network was first simulated. From this simulation, the nominal pressures H_i^0 at each node i were obtained and are reported in Table 1.

Subsequently, leak scenarios at arbitrary positions l within the network were simulated. For each leak position, three independent simulations were performed, obtaining the pressure H_i^l measured at each node i . The results are presented in Table 2. These data should be used only as a benchmark instance to evaluate algorithm performance; the proposed methodology should not depend on this specific network or these particular leak locations.

Node	Nominal Pressure
Node ₁	2.230050448
Node ₂	1.060394628
Node ₃	0.963333286
Node ₄	1.041517012

Table 1: Nominal pressures of the leak-free network.

Test	Node ₁	Node ₂	Node ₃	Node ₄
Leak at Position 1				
Test 1	1.9673	0.7977	0.7006	0.7788
Test 2	1.7892	0.6196	0.5225	0.6007
Test 3	1.7443	0.5747	0.4776	0.5558
Leak at Position 2				
Test 4	1.6396	0.4602	0.3668	0.4433
Test 5	1.6396	0.4223	0.3423	0.4123
Test 6	1.6396	0.4055	0.3310	0.3980
Leak at Position 3				
Test 7	1.6396	0.4626	0.3598	0.4426
Test 8	1.6396	0.4378	0.3147	0.4138
Test 9	1.6396	0.4368	0.3130	0.4127
Leak at Position 4				
Test 10	1.6396	0.4482	0.3531	0.4259
Test 11	1.6396	0.4442	0.3494	0.4213
Test 12	1.6396	0.4205	0.3275	0.3937
Leak at Position 5				
Test 13	1.6396	0.4070	0.3249	0.3904
Test 14	1.6396	0.4094	0.3238	0.3891
Test 15	1.6396	0.4115	0.3229	0.3879
Leak at Position 6				
Test 16	1.6396	0.4253	0.2968	0.3993
Test 17	1.6396	0.4236	0.3008	0.3972
Test 18	1.6396	0.4200	0.3089	0.3930

Table 2: Simulated pressures at the nodes for different leak scenarios. Three simulations were performed for each leak position.

References

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